

B.Tech Degree V Semester (Supplementary) Examination June 2011

ME 505 POWER PLANT ENGINEERING (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer all questions)

(8x5=40)

- I. a) Define the following terms:
(i) Plant Capacity Factor (ii) Diversity Factor
(iii) Load Factor (iv) Utility Factor (v) Demand Factor
- b) Mention the factors influencing power plant design.
- c) Illustrate the advantages of hydroelectric power plants.
- d) With a neat sketch explain the working of an axial flow compressor. Also mention its advantages.
- e) Enumerate the advantages of a nuclear power plant.
- f) What is meant by fluidized bed combustion? Explain.
- g) Mention the salient features of batch type biogas plant.
- h) With a neat sketch explain the principle of working of a Thermionic Generator.

PART B

(4x15=60)

- II. a) What are Load Curves? Explain. (7)
- b) A hydro power plant is to be used as peak load plant at an annual load factor of 30%. The electrical energy obtained during the year is 750×10^5 kWh. Determine the maximum demand. If the plant capacity factor is 24%, find reserve capacity of the plant. (8)

OR

- III. A new factory having a minimum demand of 100kW and a load factor of 25% is comparing two power supply agencies. (15)
- (a) Public supply tariff is Rs.40 per kW of maximum demand plus 2 paise per kWh. Capital cost = Rs.70,000 Interest and depreciation=10%
- (b) Private oil engine generating station.
Capital Cost = Rs.250,000 Fuel consumption =0.3kg per kWh
Cost of fuel = Rs. 70 per tone Wages = 0.4 paise per kWh
Maintenance cost = 0.3 paise per kWh Interest and depreciation =15%

- IV. Explain with the help of a suitable sketch the essential features required for a Hydraulic Power Plant. (15)

OR

- V. a) A six-cylinder two-stroke cycle marine diesel engine with 100mm bore and 120mm stroke delivers 200 B.H.P. at 2000 R.P.M and uses 100kg of fuel per hour. If I.H.P is 240, determine the following: (6)
- (i) Torque (ii) Mechanical efficiency (iii) Indicated specific fuel consumption.
- b) Explain the various types of superchargers with the help of sketches. Mention the advantages of using superchargers. (9)

(P.T.O.)

- VI. With the help of sketches explain the various types of stokers available for feeding large steam power plants. (15)
- OR**
- VII. a) What are different types of reactors commonly used in nuclear power stations? Describe the fast breeder reactor. (8)
- b) Mention the safety measures to be followed for nuclear power plants. (7)
- VIII. Write short notes on: (3x5=15)
- (i) Ocean Thermal Electric Conversion
 - (ii) Geothermal Energy
 - (iii) Magneto Hydro Dynamic power
- OR**
- IX. Explain the various ways by which solar energy is converted to useful form and used for different needs. (15)